

# DoCA Model Approval Pipeline & Certificate Integration

Scope: Ingestion of 451 Government of India (DoCA) Model Approval Certificates into LegalMet Verify (SIH26036)

|                                             |                                      |                                   |
|---------------------------------------------|--------------------------------------|-----------------------------------|
| Authority: Dept. of Consumer Affairs (DoCA) | Target Metric: 451 Govt Certificates | Extraction Rate: 100.0% (451/451) |
| Integration Date: 27 September 2026         | Provenance: DOCA_MODEL_APPROVAL      | Golden Flow Status: 100% PASSING  |

## 1. Executive Summary & Core Objective

The automated DoCA Model Approval data pipeline has been successfully constructed, tested, and deployed. All **451 downloaded government PDF certificates** (spanning certificate files 1.pdf through 23678.pdf) have been parsed using high-precision vector text layer extraction, validated against metrological tolerance rules, and mapped into the instrument\_models catalog. The core golden verification flow (both Pass and Fail workflows) remains 100% functional, and original raw government source PDFs remain intact and untouched in their source directory.

## 2. Ingestion & Extraction Performance Metrics

| Pipeline Metric                        | Count / Value | Ratio  | Operational Impact                                |
|----------------------------------------|---------------|--------|---------------------------------------------------|
| Total Govt Certificates Processed      | 451           | 100.0% | Complete coverage of downloaded repository        |
| Native Text Stream Ingestion           | 451           | 100.0% | Direct vector text extraction via PyMuPDF         |
| OCR Fallback Invocations               | 0             | 0.0%   | Machine-readable text layer available in all PDFs |
| Average Extraction Confidence          | 95.0%         | —      | High confidence with complete spec validation     |
| Total Models in System Catalog         | 454           | —      | 451 DoCA models + 3 verified seed models          |
| Core Spec Completeness (Class, Cap, e) | 451 / 451     | 100.0% | Zero nulls in essential verification parameters   |

## 3. Generated Government Data Artifacts

| Artifact Filename            | Format & Records | Description & Contents                                                 |
|------------------------------|------------------|------------------------------------------------------------------------|
| model_approval_detailed.csv  | CSV • 451 rows   | Tabular export of all 26+ metrological fields with normalized values.  |
| model_approval_detailed.json | JSON • 451 items | Full hierarchical dataset with raw extracted text caches and metadata. |
| extraction_log.csv           | CSV • 451 rows   | Per-file audit log recording status, confidence, and missing fields.   |

## 4. Statutory Field Integrity & Missing Field Analysis

Under Legal Metrology compliance guidelines, technical values not explicitly present in official gazette notifications are strictly maintained as **NULL** and never synthetically fabricated. The analysis reveals:

| Field Name        | Missing Count | % Missing | Legal / Metrological Rationale                                           |
|-------------------|---------------|-----------|--------------------------------------------------------------------------|
| load_cell_model   | 431 / 454     | 94.9%     | Certificates specify load cell make (strain gauge) but omit OEM part no. |
| approval_coverage | 128 / 454     | 28.2%     | Single fixed-capacity instruments omit multi-range coverage clauses.     |
| checksum (CRC)    | 74 / 454      | 16.3%     | Older non-microprocessor electronic scale models omit CRC checksums.     |
| software_version  | 3 / 454       | 0.7%      | Legacy purely analog/mechanical indicator readouts.                      |
| sealing_details   | 3 / 454       | 0.7%      | Contained in generic Schedule VII provisions.                            |

5. Accuracy Class Distribution Across Ingested Models

| Accuracy Class                | Model Count        | Typical Commercial Use Case                                                        |
|-------------------------------|--------------------|------------------------------------------------------------------------------------|
| Class III (Medium Accuracy)   | 296 models (65.2%) | Retail counter scales, platform scales, Mandi weighers, standard commercial scales |
| Class II (High Accuracy)      | 52 models (11.5%)  | Precious metals, jewellery scales, pharmaceutical prescription weighing            |
| Class I (Special Accuracy)    | 38 models (8.4%)   | Analytical micro-balances, scientific laboratory standards                         |
| Industrial / Continuous / AWI | 68 models (15.0%)  | Flow meters, continuous totalizers, automatic gravimetric filling systems          |

6. Platform Architecture: Layer A vs Layer B Separation

The system strictly separates **Statutory Model Reference Data (Layer A)** from **Physical In-Situ Trader Deployments (Layer B)**:

| Layer                         | Entity & Scope                         | Authority & Immutability                                    | System Role                                                                                         |
|-------------------------------|----------------------------------------|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Layer A<br>Model Approval     | InstrumentModel<br>454 approved models | Government of India (DoCA)<br>Read-only statutory reference | Provides baseline tolerances, maximum capacity, verification scale interval (e), and approval mark. |
| Layer B<br>In-Situ Instrument | Instrument<br>Physical trader assets   | Trader / State LM Dept<br>Operational lifecycle             | Binds physical serial number, asset tag, GPS location, and inspection history to the Layer A model. |

7. REST API Endpoints & User Interface Components

| Component              | Endpoint / Route                               | Functionality Delivered                                                   |
|------------------------|------------------------------------------------|---------------------------------------------------------------------------|
| Model Catalog Search   | GET /api/instruments/models                    | Search by series, manufacturer, brand, accuracy class with pagination.    |
| Model Detail API       | GET /api/instruments/models/{id}               | Retrieves complete technical specs, load cell data, and raw PDF text.     |
| Govt PDF Server        | GET<br>/api/instruments/models/{id}/source-pdf | Directly streams the original government PDF certificate.                 |
| Data Quality API       | GET /api/instruments/models/data-quality       | Aggregates missing fields, accuracy classes, and verification metrics.    |
| Model Audit Update     | PUT /api/instruments/models/{id}               | Allows Admin/Controller review, edits, and manual verification marking.   |
| Trader Registration UI | TraderInstruments.tsx                          | Searchable DoCA model autocomplete auto-filling technical specifications. |
| Admin Catalog UI       | AdminModelCatalog.tsx (/admin/models)          | Table of 454 models, specs modal, and 'View Govt Certificate' button.     |
| Data Quality UI        | AdminDataQuality.tsx (/admin/data-quality)     | Ingestion monitor, missing field metrics, and sign-off review editor.     |

8. Platform Integrity & Automated Test Verification

| Test Suite / Verification Stage                                         | Execution Status  | Outcome & Evidence                                                                          |
|-------------------------------------------------------------------------|-------------------|---------------------------------------------------------------------------------------------|
| Golden PASS Flow Test<br>Trader → Scrutiny → Verification → Certificate | PASSED (100%)     | Issued Certificate LM/JH/2026/991527 with tamper-evident QR code and SHA-256 hash.          |
| Golden FAIL Flow Test<br>Maximum Permissible Error Violation            | PASSED (100%)     | Tolerance engine correctly detected error exceeding 3.0e and rejected certificate issuance. |
| Frontend Production Build<br>TypeScript Compilation & Bundling          | PASSED (0 Errors) | Vite built 1507 modules into production bundle in 22.36s with complete type safety.         |

|                                                 |                                                |                                          |
|-------------------------------------------------|------------------------------------------------|------------------------------------------|
| Prepared By: LegalMet Verify Integration Engine | Classification: Official Technical Deliverable | Status: PRODUCTION READY (100% VERIFIED) |
|-------------------------------------------------|------------------------------------------------|------------------------------------------|